

Paper Reading: Rethinking the Value of Network Pruning (1810.05270v2)

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1. What are the motivations for this work?
2. What is the proposed solution?
3. What is the work's evaluation of the proposed solution?
4. What is your analysis of the identified problem, idea, and evaluation?
5. What are future directions for this research?
6. What questions are you left with?

1. What are the motivations for this work?

The motivation for this work stems from the widespread use of network pruning to reduce the computational cost of deep models. The authors aim to critically scrutinize the standard three-stage pruning pipeline: 1) training a large model, 2) pruning, and 3) fine-tuning.

Specifically, they challenge two common beliefs in the field: 1. That starting with a large, over-parameterized model is necessary to obtain a high-performance small model. 2. That the weights preserved after pruning are essential, and that fine-tuning is superior to training the pruned architecture from scratch.

2. What is the proposed solution?

The paper does not propose a new pruning algorithm *per se*, but rather a new **evaluation methodology** and a **change in perspective**.

- **Training from Scratch:** They propose to discard the bits of "wisdom" that suggest fine-tuning is required. Instead, they take the *architecture* resulting from a pruning algorithm and train it from random initialization ("from scratch").
- **Fair Comparison Baselines:** They introduce strict control benchmarks:
- **Scratch-E:** Training the pruned model for the same number of epochs as the large model.
- **Scratch-B:** Training the pruned model for the same computational budget (FLOPs) as the large model (typically allowing for more epochs since the model is smaller).

They apply this methodology across various state-of-the-art pruning methods, including: * **Predefined Structured Pruning** (e.g., L1-norm based filter pruning). * **Automatic Structured Pruning** (e.g., Network Slimming). * **Unstructured Pruning** (e.g., Magnitude-based pruning).

3. What is the work's evaluation of the proposed solution?

The evaluation is extensive, covering CIFAR-10, CIFAR-100, and ImageNet datasets with VGG, ResNet, and DenseNet architectures. Key findings include:

- **Structured Pruning:** For almost all structured pruning methods (both predefined and automatic), training the pruned architecture from scratch achieves accuracy **comparable to or better than** the traditional pruning + fine-tuning pipeline.
 - *Observation:* The "important" weights learned by the large model are not actually useful for the small model; what matters is the *architecture* found by the pruning process.
- **Unstructured Pruning:**
 - On small datasets (CIFAR), training from scratch works well.
 - On large datasets (ImageNet), training a highly sparse network from scratch is difficult, and fine-tuning still holds an advantage when sparsity is high.
- **Lottery Ticket Hypothesis:** The authors compare their findings with the "Lottery Ticket Hypothesis". They find that if a standard (large) learning rate is used, the "winning ticket" initialization provides no benefit over random initialization. It only helps when using suboptimal (small) learning rates.
- **Architecture Search:** They demonstrate that pruning algorithms are effectively performing **Architecture Search**. Architectures derived from pruning (e.g., via Network Slimming) are significantly more parameter-efficient than uniformly pruned networks.

4. What is your analysis of the identified problem, idea, and evaluation?

- **Problem:** The problem identification is excellent. In machine learning, it is easy for "folk wisdom" to become accepted fact without rigorous isolation of variables. By questioning the necessity of the heavy "pre-train then prune" pipeline, the authors address a fundamental efficiency question.
- **Idea:** The pivot from "pruning as compression" to "pruning as architecture search" is insightful. It shifts the focus from *weights* to *topology*. This

simplifies the deployment pipeline significantly—if you only need the architecture, you don't need to store or transfer the large model's weights.

- **Evaluation:** The use of **Scratch-B** is particularly strong. Often, small models are trained for fewer FLOPs than large ones, making comparisons unfair. By equalizing the compute budget, the authors prove that the small models have sufficient capacity to learn well on their own. The counter-evidence on ImageNet unstructured pruning adds nuance and credibility—they didn't just cherry-pick results that supported their hypothesis.

5. What are future directions for this research?

- **Pruning as NAS:** Explicitly designing algorithms that use pruning signals to discover efficient architectures which are then trained from scratch.
- **Guided Design:** Extracting "design principles" (e.g., how layer widths should vary) from pruned models and applying them to design new architectures manually, as demonstrated in their "Guided Pruning" experiments.
- **Training Sparse Networks:** Investigating why unstructured sparse networks are hard to train from scratch on ImageNet and developing optimization techniques to fix this (closely related to the later "RigL" or "Sparse evolutionary training" research).
- **Baseline Standards:** Establishing "Train from Scratch" as a mandatory baseline for all future pruning papers.

6. What questions are you left with?

- **Optimization Landscape:** Why exactly does the "winning ticket" initialization fail to outperform random initialization with large learning rates? Does the large learning rate simply allow the model to escape the local basin of the initialization immediately?
- **Domain Specificity:** Do these results hold for Recurrent Neural Networks (RNNs) or Transformers (LLMs)? (Later research suggests LLMs might behave differently regarding the value of pre-training).
- **Computational Cost of Search:** If we treat pruning as architecture search, we still have to train the large model first to find the architecture. Can we find the architecture *without* full training of the large model?

- **Transferability:** The paper shows "Transferred Guided Pruning" works between CIFAR-10 and CIFAR-100. Does a pruning pattern learned on ImageNet transfer to Object Detection (COCO) or Segmentation efficiently?